

MCQ-7

Topic: Machine Learning – Regression, OLS, Gradient Descent, Validation & Regularization

Student Name: _____ Date: _____ Score: _____

Instructions: Select the correct option by filling or marking the checkbox ☐ next to it.

1. A model is trained on emails labeled “spam” or “not spam.” What type of learning is this?
 - ☐ Unsupervised learning
 - ☐ Supervised learning
 - ☐ Reinforcement learning
 - ☐ Semi-supervised clustering
2. You’re given customer purchase data with NO labels and asked to find natural groups of similar customers. Which approach fits best?
 - ☐ Simple linear regression
 - ☐ Multiple linear regression
 - ☐ Unsupervised clustering (e.g., K-Means)
 - ☐ Gradient boosted classifier
3. In simple linear regression $y = \beta_0 + \beta_1 x + \varepsilon$, what does β_1 represent?
 - ☐ The predicted value of y when $x = 0$
 - ☐ The average change in y for a one-unit change in x
 - ☐ The variance of the residuals
 - ☐ The correlation coefficient between x and y
 - ☐ The total sum of squares
4. Suppose a multiple linear regression model predicts house price as:

$$\hat{y} = 50000 + 200x_1 + 10000x_2$$

where x_1 is square footage and x_2 is the number of bedrooms. What does the coefficient 200 represent?

- ☐ Increasing the number of bedrooms by one increases the predicted price by 200
 - ☐ Increasing square footage by one unit increases the predicted price by 200, holding the number of bedrooms constant
 - ☐ The intercept of the model is 200
 - ☐ The prediction error of the model is 200
5. In multiple linear regression, what problem arises when two independent variables are highly correlated with each other?

- ☐ Overfitting on the test set only
 - ☐ Multicollinearity, causing unstable/unreliable coefficient estimates
 - ☐ The model becomes unsupervised
 - ☐ Gradient descent stops converging entirely
6. Polynomial regression $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$ is still called “linear regression” because:
- ☐ The graph of the model is a straight line
 - ☐ It’s linear in the parameters (β ’s), even though it’s nonlinear in x
 - ☐ x^2 and x^3 are ignored during training
 - ☐ It uses the same code as simple linear regression with no changes
7. You fit a degree-15 polynomial to only 20 data points and get near-zero training error but terrible test error. What’s happening?
- ☐ Underfitting due to high bias
 - ☐ Overfitting due to excessive model complexity
 - ☐ The OLS solution doesn’t exist for polynomials
 - ☐ Gradient descent has diverged
8. The Ordinary Least Squares (OLS) technique estimates coefficients by:
- ☐ Maximizing the sum of squared residuals
 - ☐ Minimizing the sum of squared residuals between predicted and actual values
 - ☐ Minimizing the absolute value of residuals only
 - ☐ Randomly searching the parameter space until error is low
9. The closed-form OLS solution is $\beta = (X^T X)^{-1} X^T y$. This formula fails or becomes unreliable when:
- ☐ y has negative values
 - ☐ $X^T X$ is singular or near-singular (e.g., due to multicollinearity or more features than samples)
 - ☐ The learning rate is too high
 - ☐ There are more than 2 features
10. In gradient descent for linear regression, what does the learning rate (α) control?
- ☐ The number of features used in the model
 - ☐ The size of the step taken in the direction opposite the gradient at each iteration
 - ☐ Whether the model is linear or polynomial
 - ☐ The regularization strength
11. You run gradient descent and notice the cost function value is increasing every iteration instead of decreasing. What is the most likely cause?
- ☐ The learning rate is too small
 - ☐ The learning rate is too large, causing overshooting/divergence
 - ☐ You used L2 regularization
 - ☐ Too few epochs were used

12. Suppose the current parameter is $\beta = 5$, the gradient is 4, and the learning rate is $\alpha = 0.1$. Using gradient descent, what is the updated value of β ?
- ☐ 5.4
 - ☐ 4.6
 - ☐ 4.0
 - ☐ 1.0
13. L_1 regularization (Lasso) adds which penalty term to the cost function?
- ☐ $\lambda \sum_{i=1}^n \beta_i^2$ (sum of squared coefficients)
 - ☐ $\lambda \sum_{i=1}^n |\beta_i|$ (sum of absolute coefficients)
 - ☐ $\lambda \sum_{i=1}^n (\beta_i - \bar{\beta})$ (deviation from mean coefficient)
 - ☐ $\lambda \cdot \max(\beta_i)$ (largest coefficient only)
14. Which property of L_1 regularization makes it useful for feature selection?
- ☐ It always increases the magnitude of coefficients
 - ☐ It can shrink some coefficients exactly to zero
 - ☐ It forces all coefficients to have the same value
 - ☐ It removes the intercept from the model
15. You have 200 features but suspect only 15 truly matter for predicting the target. Which regularization is the most natural first choice, and why?
- ☐ L_2 (Ridge) – because it eliminates irrelevant features completely
 - ☐ L_1 (Lasso) – because it can shrink irrelevant coefficients to exactly zero, performing automatic feature selection
 - ☐ No regularization – OLS already handles this
 - ☐ Increase the polynomial degree instead
16. Why is k -fold cross-validation generally preferred over a single train-test split for evaluating a regression model?
- ☐ It trains the model faster
 - ☐ It gives a more robust estimate of generalization performance by averaging results across multiple train-test splits
 - ☐ It eliminates the need for a validation metric like RMSE
 - ☐ It guarantees the model won't overfit

17. Which validation metric penalizes large errors more heavily than small ones, making it sensitive to outliers?

- ☐ Mean Absolute Error (MAE)
- ☐ Root Mean Squared Error (RMSE)
- ☐ R^2 score alone
- ☐ Median Absolute Error

18. A regression model has $R^2 = 0.95$ on training data but $R^2 = 0.40$ on validation data. What does this gap most likely indicate, and what's a reasonable fix?

- ☐ Underfitting – add more polynomial features
- ☐ Overfitting – consider regularization (L_1/L_2), simpler model, or more training data
- ☐ The validation set is unsupervised, so R^2 is invalid
- ☐ Nothing wrong – R^2 naturally drops on unseen data

19. A Ridge regression model gives the following validation MSE for different values of α :

α	Validation MSE
0.01	120
0.1	105
1	95
10	140

Which value of α should be selected based on validation performance?

- ☐ 0.01
- ☐ 0.1
- ☐ 1
- ☐ 10

20. The actual values are $y = [2, 4, 6]$ and the predicted values are $\hat{y} = [3, 5, 4]$. What is the Mean Squared Error (MSE)?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 6